

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	1 july 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input	User enters Life Expectancy at Birth . User enters Expected Years of Schooling . User enters Mean Years of Schooling . User enters Gross National Income (GNI) per Capita .
FR-2	Input Validation	Validate that all fields are filled. Accept only valid numeric values. Display an error message for invalid or missing input.
FR-3	HDI Prediction	Send the validated input to the trained Linear Regression model. Generate the predicted HDI score.
FR-4	Development Category Classification	Classify the predicted HDI into Low , Medium , High , or Very High based on the HDI value.
FR-5	Display Prediction Result	Display the predicted HDI score. Display the corresponding development category on the result page.
FR-6	Machine Learning Model Loading	Load the trained model (HDI.pk1) when the Flask application starts. Use the model for real-time predictions.
FR-7	Web Application Interface	Provide a simple, user-friendly web interface for

		entering inputs and viewing prediction results.
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Non-Functional Requirements

FR No.	Non-Functional Requirement	Description
. NFR-1	Usability	The application provides a simple, intuitive, and user-friendly interface that allows users to enter HDI indicators and view prediction results with minimal effort.
NFR-2	Security	The system validates user inputs to prevent invalid data. It does not store any personal information or require user authentication, ensuring data privacy.
NFR-3	Reliability	The application consistently generates accurate HDI predictions using the trained Linear Regression model and performs reliably during continuous operation.
NFR-4	Performance	The system predicts the HDI score within a few seconds using minimal resources, providing fast response times for users.
NFR-5	Availability	The application is available whenever the Flask server is running and can be deployed on local or cloud platforms for continuous access.
NFR-6	Scalability	The modular architecture allows easy integration of new machine learning models, additional datasets, and future enhancements without major code modifications.